

1. HM3D-OVON: A Dataset and Benchmark for Open-Vocabulary Object Goal Navigation — 2024

Overview: HM3D-OVON is basically a benchmark designed for robots that need to find objects in indoor environments based on natural-language instructions. Unlike traditional Object Goal Navigation systems, it supports a much larger range of object categories. The robot does not actually need to depend only on a small fixed list of objects.

Project Objective: The actual and main objective is to improve a robot's ability to navigate toward objects that may not have been encountered during training. The robot should understand the object requested through natural language. It should identify the semantic meaning of the requested object. It should navigate through an unfamiliar indoor environment. The system should work even when the target object was not specifically trained beforehand. The benchmark also aims to provide a standardized way of evaluating ObjectNav systems. Ultimately, the work moves toward more flexible human-like robot navigation.

Research Gap: At first ObjectNav benchmarks generally used only a small number of predefined object categories. The systems were not suitable for open-ended real-world object searches. A user may ask for an object that is outside the robot's predefined sections or categories. Existing methods also had difficulty generalizing to unfamiliar household objects. Therefore, a larger and more diverse object-navigation benchmark was needed. The researchers also found the need for free-form language-based object goals. HM3D-OVON addresses this gap by expanding both object diversity and language-based goals.

Research Questions

1. Can robots navigate toward objects specified using free-form language?
2. Can an ObjectNav system generalize to a much larger range of object categories?
3. How well can robots search for objects they have not specifically been trained on?
4. Does open-vocabulary navigation improve flexibility compared with fixed-category navigation?
5. Can robots maintain useful navigation performance in realistic indoor environments?
6. How robust are these systems to localization and movement-related noise?
7. Can this benchmark support future development of more general object-search robots?

Key Contribution The paper introduces the **HM3D-OVON dataset and benchmark**. It provides more than 15,000 annotated household-object instances. It covers 379 different object categories. It supports free-form language as the navigation goal. It provides baseline evaluations of open-vocabulary ObjectNav approaches. The benchmark supports research into finding arbitrary objects in realistic environments. This makes it highly relevant to RoboVault's **lost-item recognition and search** component.

2. Language-Grounded Dynamic Scene Graphs for Interactive Object Search With Mobile Manipulation

Overview: This paper presents a system called **MoMa-LLM** for interactive robotic object search. It focuses on robots operating in large and initially unexplored environments. The system combines large language models with open-vocabulary scene graphs. The scene representation is continuously updated as the robot explores. Object-centric actions are connected to the robot's language-based reasoning. The system can perform zero-shot and open-vocabulary object-search tasks. Experiments were performed in both simulation and real-world environments.

Project Objective The main objective is to enable autonomous long-horizon robotic tasks. The robot should understand objects and their relationships within an environment. It should use natural-language information to reason about the search task. The robot should navigate through previously unexplored areas. It should identify relevant objects during its exploration. The system also aims to connect searching with physical manipulation. This creates a foundation for robots that can search for and interact with household objects.

Research Gap: Previous research often concentrated on either navigation or manipulation separately. Many systems also assumed that the environment had already been explored. This limits their usefulness for real-world long-horizon household tasks. Robots need to understand changing object relationships while exploring. They also need to connect language instructions with physical robot actions. Existing approaches did not fully combine these capabilities into one framework. MoMa-LLM addresses this gap through dynamic scene graphs and language-grounded actions.

Research Questions

1. Can language models help robots reason about object-search tasks?
2. Can dynamic scene graphs represent information discovered during exploration?
3. Can robots search for objects in environments they have not previously explored?
4. Can open-vocabulary recognition support different types of household objects?
5. Can language reasoning be connected directly to object-centric robot actions?
6. Does combining navigation and manipulation improve search efficiency?
7. Can the approach transfer from simulated environments to real robots?

Key Contribution

The paper introduces the **MoMa-LLM** framework. It combines LLM reasoning with open-vocabulary dynamic scene graphs. The scene representation is updated as the robot explores the environment. It provides an object-centric action space for mobile manipulation. The approach supports zero-shot and open-vocabulary object search. Experiments demonstrate improved search efficiency over conventional approaches. The work provides a strong foundation for RoboVault's **search-and-retrieval architecture**.

3. VLA: Exploration and Exploitation Based on Visual-Language Aligned Information for Robotic Object Goal Navigation

Overview

VLAI focuses on improving robotic Object Goal Navigation in unfamiliar indoor environments. The robot needs to locate a specific target object while it explores an environment that it does not completely know. The approach combines visual information with language-based semantic information. The system examines objects and information available around unexplored areas of the environment. It uses a language-based description system to understand the semantic characteristics of these areas. The similarity between the target object and surrounding objects is then used to guide the robot. This allows the robot to make more informed decisions about where it should search next.

Project Objective

The primary objective of VLAI is to make robotic object searching more efficient. Instead of allowing the robot to explore every possible location equally, the system attempts to identify locations that are more likely to contain the target. It aims to use semantic relationships between the requested object and other visible objects. For example, information about tables, shelves, beds, or desks may help the robot determine where a particular target could be located. The system also attempts to balance exploration of unknown areas with exploitation of promising areas. This reduces unnecessary navigation and search time. Ultimately, the objective is to provide robots with a more intelligent strategy for finding objects in unfamiliar environments.

Research Gap

Traditional navigation methods often rely heavily on geometric information such as distances, obstacles, and map structure. However, geometric information alone does not tell a robot where an object is likely to be found. For example, knowing that a table exists is more useful for finding a laptop than simply knowing the distance to another room. Previous approaches therefore lacked sufficient semantic understanding of the environment. Robots could spend considerable time exploring areas that had little connection with the target object. There was a need for methods that combine visual perception with language-level semantic reasoning. VLAI addresses this gap by using visual-language information to guide exploration and target selection.

Research Questions

1. Can visual-language information improve robotic ObjectNav?
2. Can descriptions of unexplored areas help a robot choose better destinations?
3. Can semantic similarity between objects guide the robot toward the target?
4. Can previous visual observations improve global environmental understanding?
5. Does combining language and vision reduce inefficient exploration?
6. Can the approach improve navigation performance in unseen indoor environments?
7. Can visual-language reasoning make object-search robots more intelligent?

Key Contribution

The key contribution is a visual-language aligned exploration strategy for robotic ObjectNav. The system uses a Chat Descriptor to generate semantic descriptions of objects around unexplored areas. These descriptions are combined with semantic similarity information related to the target object. The resulting information helps the robot select more promising long-term navigation goals. The approach therefore combines visual perception with language-based semantic reasoning. Experiments using indoor environments such as Gibson and HM3D demonstrate the effectiveness of the approach. For RoboVault, this research can support an **intelligent search strategy that decides which room or area should be searched first**.

4. OpenIN: Open-Vocabulary Instance-Oriented Navigation in Dynamic Domestic Environments — 2025

Overview

OpenIN focuses on autonomous navigation toward specific object instances in dynamic domestic environments. Real household environments are constantly changing because objects can be moved, placed inside containers, or carried to different locations. Therefore, simply recognizing an object category is not always sufficient. OpenIN uses open-vocabulary recognition to identify a wider variety of objects. It also introduces a Carrier-Relationship Scene Graph to represent relationships between objects and their carriers or locations. The robot updates this information as it observes changes in the environment. This allows the system to reason about object locations more effectively during navigation.

Project Objective

The main objective of OpenIN is to allow robots to find particular object instances in changing domestic environments. The robot should not only understand the category of the requested object but also identify the relevant instance. The system aims to maintain information about where objects are located and how they are related to other objects. It should update this information when an object is moved. Another objective is to use language and visual information to improve target selection. The robot should therefore be able to adapt its navigation decisions as the environment changes. These capabilities are important for realistic autonomous object-search applications.

Research Gap

Many existing ObjectNav systems work well in environments where objects remain relatively stationary. However, real homes are dynamic, and objects can frequently change their positions. Another limitation is that category-level recognition does not distinguish between multiple instances of the same object. For example, identifying a “cup” does not necessarily tell the robot which cup belongs to the user. Static maps also cannot accurately represent continuously changing object relationships. Therefore, robots require dynamic semantic representations. OpenIN addresses these gaps by combining open-vocabulary recognition with dynamic object-carrier relationships.

Research Questions

1. How can robots locate a specific object instance in a dynamic environment?
2. How can a robot recognize objects that are not restricted to predefined categories?
3. How should the robot update its knowledge when an object moves?
4. Can object-carrier relationships help predict where a moved object may be located?
5. Can language-based commonsense knowledge improve instance navigation?
6. Can visual-language similarity improve target selection?
7. Can the proposed approach operate effectively on a real robot?

Key Contribution

The main contribution is the Carrier-Relationship Scene Graph developed for dynamic object navigation. This representation allows the robot to maintain relationships between objects and the things or locations associated with them. The system updates these relationships as the robot observes changes in its environment. Open-vocabulary information allows the robot to deal with a wider range of target objects. Language-based commonsense knowledge is also used to support navigation decisions. The approach is evaluated in both simulated and real-world environments. For RoboVault, this paper is particularly important because a **lost item may have been moved from its original location**, requiring the robot to continuously update its search strategy.

5. One Map to Find Them All: Real-time Open-Vocabulary Mapping for Zero-shot Multi-Object Navigation

Overview

OneMap presents a real-time open-vocabulary mapping system for zero-shot multi-object navigation. The main idea is that a robot should remember useful information about its environment rather than starting every object search from the beginning. The system creates a semantic map containing information about objects and their spatial locations. This map can be reused when the robot receives another object-search request. The approach supports objects that were not necessarily included in a fixed training category list. The researchers evaluate the method in simulation and on a physical robot. This makes the research particularly relevant to real-world robots that need to perform repeated searches.

Project Objective

The main objective is to improve the efficiency of multiple object searches. A robot operating in a building may receive several requests to locate different objects. Rebuilding environmental knowledge for every request would waste time and computational resources. OneMap therefore aims to create a reusable semantic representation of the environment. The robot can store information about previously observed objects and their locations. It can then use this information to make better decisions during future searches. The approach also aims to support real-time operation on robotic hardware. Overall, the project aims to provide robots with a form of persistent spatial-semantic memory.

Research Gap

Many zero-shot object-navigation approaches treat each new search task as a separate problem. As a result, information collected during earlier searches may not be effectively reused. Humans naturally remember where objects and rooms are located after exploring a building. Robots need similar capabilities to efficiently perform repeated searches. Another challenge is that visual-semantic information is not always perfectly reliable. A robot therefore needs to represent uncertainty in its semantic map. Existing approaches did not fully address reusable open-vocabulary mapping and semantic uncertainty together. OneMap attempts to fill this gap through a real-time reusable semantic mapping approach.

Research Questions

1. Can a robot reuse information from previous object searches?
2. Can a reusable semantic map improve multi-object navigation?
3. How can semantic uncertainty be represented during object searching?
4. Can previously observed objects help the robot search for new objects?
5. Can the system operate in real time on a physical robot?
6. Does memory improve both single-object and multi-object navigation?
7. Can open-vocabulary mapping provide a practical foundation for real-world object-search robots?

Key Contribution

The major contribution is the development of the OneMap real-time open-vocabulary mapping approach. It provides a reusable semantic representation that can retain information from previous searches. The system supports zero-shot navigation toward multiple different objects. It also considers uncertainty when updating semantic information in the map. The research demonstrates that previously collected information can be used to guide later searches. Importantly, the approach was evaluated on both simulated environments and a physical robot. For RoboVault, OneMap can provide the basis for a **memory system that remembers previously detected objects and their approximate locations**.

6. Object Search Strategy for Service Robots With Knowledge-Based Viewpoint Selection and Hierarchical Action Decisions — 2026

Overview

This 2026 research focuses on autonomous object searching for service robots operating in complex indoor environments. The paper considers situations where the target object's location is unknown and may change over time. It proposes a framework that combines mapping, viewpoint selection, planning, and hierarchical decision-making. A visual-topological map is used to represent information about the environment. The system estimates which areas are more likely to contain the requested object. It then selects viewpoints that can provide useful information for continuing the search. This creates a more adaptive search process compared with following a fixed route.

Project Objective

The primary objective is to make service robots more efficient when searching for objects. The robot should be able to determine which areas have a higher probability of containing the target. It should then select useful viewpoints rather than moving randomly through the environment. Another objective is to minimize unnecessary movement and reduce search time. The robot should continuously update its knowledge based on new observations. The framework also aims to combine high-level global planning with low-level local exploration. Overall, the objective is to create an autonomous search strategy that can adapt to complex and changing indoor environments.

Research Gap

Object searching remains challenging when a robot does not know where the target is located. Fixed search patterns can result in considerable unnecessary movement. The problem becomes more difficult when the target object changes position or when the robot's initial assumptions are incorrect. Robots therefore need to combine previous knowledge with real-time sensory information. They also need to determine which viewpoint will provide the most useful

information. Existing systems may not sufficiently combine global planning, local exploration, and knowledge-based viewpoint selection. This research addresses these limitations by integrating multiple planning and decision-making mechanisms into one object-search framework.

Research Questions

1. How can a service robot select the most useful viewpoint for finding an object?
2. Can prior knowledge about rooms and object relationships improve search efficiency?
3. How can a robot update its search probabilities during exploration?
4. Can global and local navigation strategies be combined effectively?
5. How should the robot react when its initial search assumptions are incorrect?
6. Can spatial memory help the robot make better exploration decisions?
7. Can hierarchical planning improve object search in complex indoor environments

Key Contribution

The paper proposes a unified framework for object search and navigation. It introduces a **visual-topological map (VTMap)** for spatial and semantic reasoning. Prior knowledge and object co-occurrence statistics are used to estimate target locations. Gaussian Mixture Models help construct room-level probability distributions. The framework combines global planning with adaptive local exploration. It also incorporates action-cost modeling and hierarchical decision-making. RoboVault can apply these ideas to choose **which room or location to search next for a lost item**.

7. Zero-shot Object Navigation Method Based on Open-vocabulary Object Detection — 2026

Overview

This 2026 paper proposes a zero-shot Object Navigation approach based on open-vocabulary object detection. The system is designed for robots that need to search for objects that may not have been explicitly included in their training categories. It focuses on improving both object-recognition accuracy and navigation efficiency. The method introduces spatial constraints to improve object detection during navigation. It also uses confidence fusion to make detection decisions more reliable. Target-oriented semantic information is extracted to help the robot

understand the relationship between the requested object and the observed environment. The research therefore connects open-vocabulary computer vision with autonomous robot navigation.

Project Objective

The main objective is to allow robots to navigate toward unfamiliar objects without requiring separate training for every possible object category. The system aims to make open-vocabulary object recognition more suitable for real-time robotic applications. It attempts to reduce computational requirements while maintaining accurate target detection. Another objective is to use spatial information to reduce false object detections. The system also aims to improve the robot's understanding of target-related environmental information. By combining perception and navigation, the robot can make better decisions about where to move. The overall goal is a more flexible and practical zero-shot object-navigation system.

Research Gap

Traditional object-detection systems usually operate using a predefined set of categories. This becomes a major limitation when a user asks a robot to find an unfamiliar object. Some existing zero-shot methods also require significant computational resources. Large language and vision models can introduce additional processing delays. Another problem is that object recognition can become unreliable when objects are visually similar or partially hidden. Robots also need to understand the spatial relationship between detected objects and the navigation target. This research attempts to address these gaps through open-vocabulary detection, spatial constraints, confidence fusion, and target-oriented semantic processing.

Research Questions

1. Can open-vocabulary detection support real-time zero-shot object navigation?
2. How can object-detection accuracy be improved without excessive computational cost?
3. Can spatial constraints improve the reliability of target detection?
4. Can confidence fusion reduce incorrect object predictions?
5. How can target semantics be connected with environmental visual features?
6. Can the robot reduce ineffective exploration through target-oriented perception?
7. Can the proposed method generalize effectively to real-world environments?

Key Contribution

The research proposes an open-vocabulary zero-shot ObjectNav framework. It introduces spatial constraints and confidence fusion for object detection. It develops a target-oriented semantic extraction network. Cross-attention is used to connect target semantics with environmental features. The approach aims to reduce ineffective exploration and improve target recognition. Experiments report improvements over TDANet for both unknown and known object searches. This is directly useful for RoboVault's **AI camera/object-recognition and autonomous navigation modules**.

8. Hierarchical Semantic Knowledge-Based Object Search Method for Household Robots

IEEE Transactions on Emerging Topics in Computational Intelligence

Overview

This paper focuses on helping household robots search for objects efficiently. It uses semantic relationships between rooms, environmental objects, and target objects.

The robot uses this knowledge to identify likely locations of missing objects. It also considers room layout and distance during searching. The method was tested in both simulated and real environments.

Project Objective

The main objective is to reduce the time required to find objects. The system guides the robot toward rooms where the target is more likely to exist. It uses semantic knowledge to improve search decisions. The method also aims to reduce unnecessary robot movement. This is useful for an autonomous lost-and-found robot.

Research Gap

Traditional object-search methods may search large areas unnecessarily. They may not use relationships between objects and their surrounding rooms.

Dynamic home environments can also make object locations uncertain. This paper addresses these limitations through hierarchical semantic knowledge. However, it mainly focuses on household environments rather than complete lost-and-found services.

Research Questions:

1. How can semantic relationships help robots locate missing objects?
2. How can the robot select the most promising room?
3. How can search time and travel distance be reduced?
4. How can semantic knowledge be updated during searching?
5. How well does the method work in real environments?

Key Contribution:

The paper introduces a hierarchical semantic knowledge-based search method. It models relationships between rooms, environmental objects, and dynamic objects. A heuristic search strategy selects promising locations. Room layout and robot-to-room distance are also considered. Experiments show reduced search time and shorter paths.

9.Interactive Semantic Map Representation for Skill-Based Visual Object Navigation — 2024

IEEE

Overview

This paper studies visual object navigation for mobile robots. It develops a semantic map representing objects and their locations. The robot continuously builds this map while interacting with an indoor environment.

It combines learned robot skills with traditional map-based planning. The approach was evaluated using the Habitat environment.

Project Objective

The objective is to help robots navigate toward specific objects. The system identifies semantic information from visual observations. It creates intermediate exploration goals before reaching the target. This helps the robot make better navigation decisions. The approach is relevant to robots searching for lost belongings.

Research Gap

Robots need accurate semantic maps to find objects in unknown environments.

Existing approaches can struggle to maintain useful semantic information during navigation.

They may also have difficulty connecting exploration with the final object goal.

This work combines semantic mapping, exploration, and navigation skills.

However, it does not specifically address lost-and-found item management.

Research Questions:

1. How can semantic maps improve object navigation?
2. How can robots select useful intermediate exploration goals?
3. How can visual information improve navigation decisions?
4. How can learned skills work with traditional path planning?
5. How effectively can the robot reach target objects?

Key Contribution:

The paper introduces an interactive semantic-map representation.

It supports both intermediate exploration and final object navigation.

The proposed SkillTron approach combines learned skills with map planning.

Experiments demonstrate improved navigation performance.

This provides a useful foundation for RoboVault's object-search module.

10.Find Everything: A General Vision Language Model Approach to Multi-Object Search — 2025

IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)

Overview:

This paper addresses the problem of finding multiple objects in unknown environments. It proposes a system called **Finder** using vision-language models. The system creates score maps to reason about multiple target objects. It combines scene-level and object-level semantic relationships. Experiments were conducted in both simulated and real-world environments.

Project Objective:

The main objective is to efficiently search for several objects. The robot must decide where to move while considering multiple targets. Vision-language models help understand object and scene relationships. The system attempts to reduce unnecessary navigation. This is highly relevant to finding multiple lost items.

Research Gap:

Multi-object search is difficult because several targets may exist in different locations. Traditional methods may not effectively use semantic relationships between objects. They can also require extensive task-specific training. This work uses vision-language reasoning to improve generalization. However, it does not focus specifically on item ownership or return management.

Research Questions:

1. How can vision-language models improve multi-object search?
2. How can multiple target objects be tracked simultaneously?
3. How can scene and object relationships guide navigation?
4. How can travel cost be reduced during multi-object searching?
5. How well does the system perform in real environments?

Key Contribution:

The paper introduces the **Finder** approach for multi-object search. It uses multi-channel score maps to track different target objects. Scene-level and object-level semantic information are combined. The system was evaluated in simulation and real-world environments. The work provides useful concepts for RoboVault's multi-item search capability.